

AgentFail: A Diagnostic Benchmark for Silent Failures in Data-Science LLM Agents

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ABSTRACT

Large language model (LLM) agents increasingly solve data-science tasks by writing and executing code, yet aggregate success rates conceal where and how they fail. We introduce **AgentFail**, a diagnostic framework with five failure stages—Analytical-Plan, Code-Generation, Runtime, Output-Mismatch, and Answer-Error—and an independent observability label that distinguishes silent from explicitly visible failures. We evaluate five LLMs over 5,984 runs on 318 tasks from synthetic, UCI, and DSBench suites. After separating 2,218 infrastructure failures, we analyse 3,602 valid runs and release a 150-trace human-adjudicated gold set. Its observed stages are Output-Mismatch (77), Runtime (39), Answer-Error (22), and Code-Generation (12); Analytical-Plan has no final cases. On the newly annotated 103-trace tranche, raw pre-adjudication agreement is Fleiss’ $\kappa = .350$ for stage and $.590$ for observability, exposing a difficult Output-Mismatch/Analytical-Plan boundary that we resolve by consensus adjudication. Our revised oracle-assisted rule classifier obtains 91.3% accuracy (95% bootstrap CI [86.7, 95.3]) and 90.3% macro-F1 over the four observed classes; observability accuracy is 99.3%. Silent-failure rates remain task-dependent (86% synthetic, 4% UCI, 19% DSBench). A six-condition recovery benchmark yields 15.7% recovery overall (10.4% silent versus 36.7% loud), while oracle repair matches its no-op control (64%). We release code, traces, adjudications, splits, and reproducible baselines.

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1 INTRODUCTION

Large language models (LLMs) have rapidly become the reasoning core of autonomous agents that write and execute code to solve data-science tasks [8, 12]. Recent benchmarks such as DSBench [5], DataSciBench [23], and DSAEval [17] report that even the strongest agents solve only 34–88% of data-analysis tasks, leaving a large residue of failures. Yet these benchmarks report a single aggregate

number—success rate—that conceals the *structure* of failure: an agent that crashes on every hard task and an agent that silently returns plausible-but-wrong answers on the same tasks are indistinguishable under aggregate accuracy.

The problem is acute because silent failures are invisible to the agent itself. When generated code raises an exception, the agent observes the traceback and can retry; when the code executes successfully but produces a wrong answer, the agent has no signal that anything went wrong. Recent work has begun to recognise this distinction: Mehta [9] report “confident and wrong” semantic failures in coding agents, Wu [21] document silent failures in production agent runtimes, and Liu [7] frame silent failure as an entropy-driven inevitability. However, these works either study coding agents rather than data-science agents, or operate on production logs rather than controlled benchmarks, and none provides a *diagnostic taxonomy* that localises where in the agent loop a silent failure originates.

We address this gap with **AgentFail**, a diagnostic framework that assigns failures to five stages—*Analytical-Plan*, *Code-Generation*, *Runtime*, *Output-Mismatch*, and *Answer-Error*—and records observability as an independent label. This separation permits an Output-Mismatch to be silent or explicitly visible, rather than inferring observability from stage. To test repairability, oracle-guided repair replaces the originating step with a ground-truth action and re-executes; a no-op control prevents us from misrepresenting repairability as causal attribution.

We evaluate five frontier LLMs across 5,984 runs on three task suites: 95 synthetic trap tasks, 23 real UCI datasets, and 200 real DSBench financial-analysis tasks. All tables are auto-generated from a single source-of-truth manifest with assertion checks. Our analysis surfaces four findings:

- (1) **Task-dependent failure bifurcation.** On synthetic trap tasks, the corrected silent-failure rate (SFR) reaches 86%; on UCI real data it drops to 4%, and on DSBench to 19%. We report $R^2 = 0.493$ between execution-success rate and SFR as a descriptive statistic, noting that this correlation is partially mechanical (the two variables share the loud-failure count) and cannot be decomposed into “mechanical” vs. “real” components.
- (2) **Recovery conditions.** The recovery rate is 0% under the standard no-feedback protocol, but our six-condition benchmark reveals 15.7% recovery with feedback signals (10.4% for silent vs. 36.7% for loud failures).
- (3) **Oracle repairability.** Counterfactual repair matches the no-op baseline (64%), providing an upper bound on single-step repairability but not step-level causal attribution.
- (4) **Verifier boundary conditions.** A failure-aware verifier significantly helps Qwen3-max (+24.9%, McNemar $p < 0.001$) and

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GPT-4o (+19.0%, $p < 0.001$), but not other models. The benefit to Qwen3-max may partly reflect format/protocol fix-up.

Our contributions are: (i) a five-stage taxonomy and orthogonal observability label, with a final 150-trace human gold set containing Output-Mismatch (77), Runtime (39), Answer-Error (22), and Code-Generation (12); (ii) a revised oracle-assisted classifier achieving 91.3% accuracy and 90.3% macro-F1 over the four observed classes; (iii) oracle repair with no-op controls and a six-condition recovery benchmark; and (iv) 5,984 runs across 318 tasks, with a 32-field manifest that separates infrastructure failures (38.1%) from model failures.

2 RELATED WORK

2.1 Data-Science Agent Benchmarks

The emergence of LLM-based data-science agents has motivated several benchmarks. DSBench [5] collects 466 data-analysis and 74 data-modelling tasks from ModelOff and Kaggle, reporting that the best agent (GPT-4o) solves only 34% of analysis tasks. DataSciBench [23] evaluates LLMs across the full data-science workflow with multi-dimensional metrics. DSAEval [17] covers real-world data-science problems spanning multiple stages. LongDSBench [22] specifically studies long-horizon agentic data analysis and shows that agents fail on iterative tasks. GRACE-DS [18] introduces a guarded reward-guided correction environment for AutoML agents. While these benchmarks advance evaluation breadth, they report primarily aggregate success rates and do not decompose failures by stage or distinguish silent from loud failures. Moghadasi and Ghaderi [10] audit twelve agent-benchmark papers and find systematic under-reporting of evaluation details, motivating our fine-grained diagnostic metrics. Agent Laboratory [14] uses LLM agents as research assistants but does not diagnose failure modes. Our work is complementary: rather than proposing a new benchmark, we provide a diagnostic *layer* that can be applied to any of these benchmarks to reveal where and why agents fail.

2.2 LLM Agent Failure Analysis

A recent wave of work studies how and why LLM agents fail. Albayaydh et al. [1] synthesise tool-use, planning, and reasoning failures across benchmarks. ToolFailBench [16] diagnoses tool-use failures by decomposing the tool-call pipeline. Wu [21] present a longitudinal taxonomy of silent failures in a production LLM agent runtime, and Liu [7] frame silent failure as an entropy-driven inevitability of autonomous agents. Mehta [9] identify “confident and wrong” semantic failures in coding agents, the closest conceptual precursor to our silent-failure notion, but study coding rather than data-science tasks and do not provide a stage-level taxonomy. TrajAudit [19] automates failure diagnosis for agentic coding systems via trajectory auditing. Reddy et al. [13] recover a silent policy-violation failure mode in tool-using agents via deterministic gates. Zhang et al. [24] diagnose library drift as a silent failure in self-evolving skill libraries. Ekka [3] diagnoses silent errors in LLM *inference* (serving), a different layer than agent-level silent failures. Our contribution relative to this body of work is a *five-stage taxonomy* that localises silent failures within the agent loop and an *oracle-guided repair* module that estimates the reparability of failed traces.

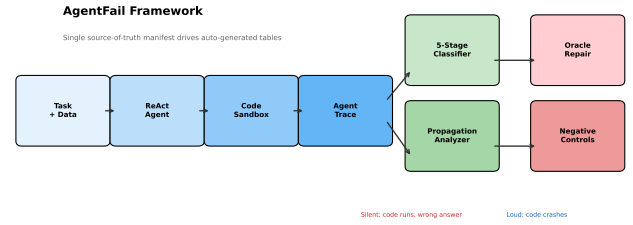


Figure 1: AgentFail framework. The agent produces a step-by-step trace; the classifier assigns each failure to one of five stages and marks it silent or loud; the propagation analyser computes how far the failure spread; causal replay replaces the originating step with the ground-truth action to test counterfactual dependence.

2.3 Causal Attribution and Failure Recovery

Shah [15] proposes Causal Agent Replay for counterfactual attribution of LLM-agent failures, which inspires our repair module; we adapt the counterfactual principle to the data-science setting and integrate it with the five-stage taxonomy. However, as our negative-control experiment shows (Section 5.3), the oracle repair rate matches a no-op baseline, so we report it as an upper bound on reparability rather than as evidence of step-level causal attribution. On the recovery side, self-consistency [20] samples multiple reasoning paths and votes, but targets reasoning errors rather than code-execution silent failures. Verification-based approaches [4] use verifier or critic agents to suppress hallucinations, but Itkin [4] shows that delayed verification can destabilise multi-agent belief. Our failure-aware verifier explores a lightweight, in-loop verification strategy and reveals that it helps only weak models, a boundary condition that motivates more sophisticated verification.

2.4 LLM-as-Judge and Evaluation Methodology

Using LLMs to evaluate LLM outputs is common [2, 6], but LLM-as-judge introduces its own biases. Mohammadi et al. [11] survey agent evaluation. We instead score against 150 human-adjudicated traces. Raw agreement on the newly annotated 103 traces is $\kappa = .350$ for stage and $.590$ for observability, and the revised oracle-assisted rule classifier obtains 91.3% accuracy and 90.3% macro-F1 over observed classes (Section 6).

3 THE AGENTFAIL FRAMEWORK

AgentFail instruments a ReAct-style data-science agent with three diagnostic layers: (1) a five-stage taxonomy plus an independent observability label; (2) a propagation-depth analyser; and (3) counterfactual replay for reparability testing. Figure 1 summarises the architecture.

3.1 Agent Architecture and Trace Recording

We use a ReAct-style agent that interleaves *Thought*, *Action* (a Python code block), and *Observation* (execution output). The agent operates within a restricted code sandbox that executes generated

Python with a persistent namespace and a working directory containing the task’s data file (CSV or XLSX). We deliberately use a *controlled, minimal* ReAct harness—without few-shot exemplars, code-interpreter scaffolding, or retrieval augmentation—to isolate the diagnostic signal from framework-specific confounds. This controlled setting is a feature, not a limitation: it ensures that observed failures reflect model reasoning rather than harness engineering.

Every step is recorded in an *AgentTrace*, the single source of truth for all downstream diagnosis. Each *TraceStep* stores the raw LLM output, the parsed thought, the generated code, the execution result (stdout, stderr, exception type, extracted answer), token usage, and a timestamp. Unlike DSbench [5], which retains only a final response, our trace records the full step-by-step trajectory, enabling stage-level localisation and propagation analysis.

3.2 Five-Stage Failure Taxonomy

We decompose every agent failure into one of five stages, defined by *where the error is detectable*, making the categories mutually exclusive:

- **Analytical-Plan.** The agent selects the wrong analytical approach: e.g., computing the overall mean when the task requires a per-group sum, or using future data in a time-series analysis (temporal leakage). Often silent.
- **Code-Generation.** The agent writes code that embodies a wrong operation: e.g., using `.count()` where `.sum()` is needed, selecting the wrong column, or introducing type confusion. Silence varies.
- **Runtime (loud).** The generated code raises an exception: runtime errors, type errors, key errors, or sandbox security blocks. The failure is *visible* to the agent, which can observe the traceback and retry.
- **Output-Mismatch.** The code executes without error but the agent extracts or reports a result unsupported by the relevant execution output. The mismatch may be silent or explicitly observable (e.g., “No data found”).
- **Answer-Error (silent).** The code runs and produces a result, but the result itself is wrong relative to the ground truth: e.g., a wrong aggregation result or an incomplete analysis. The failure is *invisible* to the agent.

Stage and observability answer different questions: stage identifies where the error becomes detectable, whereas `is_silent` records whether the trace explicitly exposes failure. Runtime is ordinarily loud and Answer-Error ordinarily silent, but Output-Mismatch and Code-Generation vary. Table 1 lists the hierarchy.

Disambiguation rule. To ensure mutual exclusivity, the classifier assigns the *first* stage at which the error becomes detectable. Consider a task requiring per-group sum where the agent uses `.count()`: the error is *detectable at code-generation time* (the code uses the wrong operation), so it is classified as Code-Generation, not Answer-Error (where the code uses the correct operation but the data produces a wrong result) or Output-Mismatch (where the code and result are correct but the agent misreports). The key distinction is: Code-Generation means the code itself is wrong; Answer-Error means the code is correct but the answer is wrong due to data issues; Output-Mismatch means the code and result are correct but

Table 1: Five-stage failure taxonomy (v2). Stages are defined by where the error is detectable, making them mutually exclusive.

Stage	Sub-categories	Observability
Analytical-Plan	wrong_operation_plan, leakage_plan	often
Code-Generation	wrong_aggregation_code, wrong_column, type_confusion	varies
Runtime	runtime_exception, key_error, security_block	No
Output-Mismatch	misread_output, hallucinated_answer, no_answer_printed	varies
Answer-Error	wrong_result, incomplete_analysis	Yes

the agent misreads or hallucinates the final answer. This ordering prevents the ambiguity where a single failure could be assigned to multiple stages.

3.3 Failure Classification

The classifier combines global exception checks, conservative Thought analysis, Thought-Code-gt_path consistency rules, whole-trace output interpretation, and explicit failure-message detection. Code-Generation rules cover wrong aggregation, wrong dynamic column selection, omitted implementation, and computed-but-unprinted answers. Because reference paths are inputs, this classifier is oracle-assisted rather than deployment-ready. On the final 150-trace human gold set it achieves 91.3% accuracy (95% bootstrap CI [86.7, 95.3]) and 90.3% macro-F1 over the four observed classes; observability accuracy is 99.3%. Code-Generation precision and recall are both 91.7%.

3.4 Propagation Depth

Once a failure originates at step i , it may propagate to later steps. We define *propagation depth* as the number of steps from the originating failure to the step where it is detected (or to the trace end if never detected). Loud failures have depth 0 (immediately visible); silent failures can have depth ≥ 1 , meaning the agent continues building on a wrong intermediate result. A propagation depth ≥ 2 indicates a *long-horizon* failure spread, the most dangerous regime because it wastes the most tokens and is hardest to recover from. Across all 3,940 failures, the average propagation depth is 0.30 steps; 88.6% are detected immediately (depth 0), while 6.9% propagate to depth ≥ 2 (see Table 11 in Appendix A).

3.5 Oracle-Guided Repair

To test whether a failed trace is *repairable*, we adapt the counterfactual principle [15] to the data-science setting. Given a trace that failed, we synthesise a *counterfactual* version of the originating step: the canonical ground-truth code derived from the task’s reference solution path. We then replay execution from that point in a fresh sandbox. If the replayed trace now produces the correct answer, the failure was repairable; if the replay still fails, the failure has a deeper root cause.

Formally, let $T = (s_1, \dots, s_n)$ be a failed trace and s_k the classified originating step. Let s_k^* be the ground-truth action for step k . We construct the counterfactual trace $T' = (s_1, \dots, s_{k-1}, s_k^*, \dots)$ and execute it. The repair is successful iff T' yields the correct answer. The *oracle repair rate* is the proportion of failed traces for which repair succeeds.

Table 2: Task suite summary.

Suite	Tasks	Runs	Failures
Synthetic trap	95	2,850	1,485
UCI real data	23	345	52
DSBench real	200	1,964	1,803
Supplementary	—	825	600
Total	318	5,984	3,940

Negative control. A no-op control runs the ground-truth code independently without any step replacement. If the oracle repair rate matches the no-op rate (as we observe, Section 5.3), the oracle rate measures only whether the ground-truth code produces the correct answer, not step-level causality. We therefore report oracle repair as an upper bound on reparability.

3.6 Diagnostic Metrics

We report three layers of metrics: **Layer 1 (task-level)**: success rate (SR), code-execution success rate. **Layer 2 (failure-diagnostic)**: silent-failure rate (SFR = proportion of failures that are silent), stage distribution, propagation depth, recovery rate, and oracle repair rate. **Layer 3 (token-economic)**: tokens-per-success and cost-per-success, addressing the under-reporting of cost identified by Moghadasi and Ghaderi [10]. Verifier comparisons use the McNemar test (b, c, χ^2 p -value).

4 EXPERIMENTS AND RESULTS

4.1 Experimental Setup

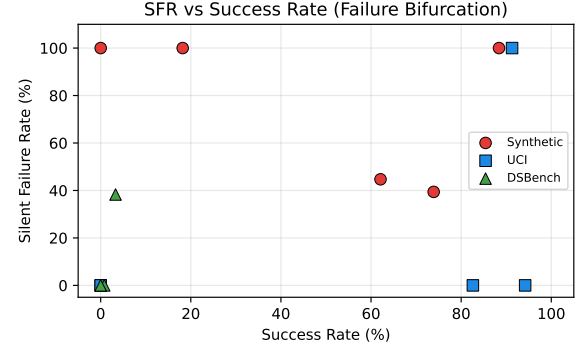
Models. We evaluate five frontier LLMs via an OpenAI-compatible endpoint (ChatAnywhere): GPT-4o, GPT-4o-mini, DeepSeek-V3 (deepseek-chat), DeepSeek-R1 (deepseek-reasoner), and Qwen3-max (qwen3-max-2026-01-23). Note that qwen3-max-2026-01-23 is a pinned version snapshot, while the other four are rolling aliases that may be updated by the provider. All models use temperature 0 and a maximum of 6–8 ReAct steps. API calls were made during June–July 2026.

Task suites. We use three complementary task suites (Table 2):

Protocol and scale. Each (model, task) pair is run for 3 repetitions at temperature 0. The primary release contains **5,984 runs** over **318 tasks**, with 2,218 infrastructure failures (38.1%) separated from 3,602 valid runs. A follow-up relaxed-sandbox collection adds 927 complete runs from GPT-4o, GPT-4o-mini, and DeepSeek-chat with no API failures, enabling full-trace human sampling. The final validation resource contains 150 human-adjudicated failed traces. Tables derive from versioned manifests with assertion checks.

Repetition agreement. Under temperature 0, 89.9% of (task, model, variant) groups show full agreement across 3 repetitions (all correct or all incorrect). The 10.1% disagreement rate reflects server-side non-determinism in the API endpoint, not sampling randomness. DeepSeek-R1 shows 100% agreement (all infrastructure failures), while GPT-4o-mini shows 84.6%.

Statistical tests. Verifier comparisons use the McNemar test (appropriate for paired binary outcomes), reporting b (baseline correct & verifier wrong), c (baseline wrong & verifier correct), and the χ^2 p -value.

**Figure 2: SFR vs. success rate across task suites and models.**

4.2 Result 1: Comprehensive Model Comparison

Table 3 presents the comprehensive baseline results across all five models, three task suites, and nine diagnostic metrics; it is the single source of truth for all success-rate and SFR figures reported in this paper, and every cell is computed over valid runs only (infrastructure failures excluded, marked “—” where no valid runs exist). Three patterns emerge. First, **DeepSeek-V3 is the strongest model** on both synthetic (88.4% SR) and UCI (94.2% SR) tasks, achieving perfect code-execution success on synthetic traps. Second, **the silent-failure rate is strongly task-dependent**: on synthetic trap tasks, SFR ranges from 39% to 100% (failures are often wrong answers, not crashes); on UCI real data, SFR is bimodal—either 0% (GPT-4o-mini, DeepSeek-V3) or 100% (GPT-4o); on DSBench, only GPT-4o produces silent failures (38.3%). Third, **DSBench is extremely challenging** for all models (0–3.3% SR), with GPT-4o achieving the highest success rate and the only non-zero SFR, indicating that even the strongest model produces plausible-but-wrong answers on complex financial data. We deliberately report SFR only for model-suite cells that have valid runs; cells consisting entirely of infrastructure failures are left undefined rather than reported as 0% or 100%.

4.3 Result 2: Failure Bifurcation

Table 4 reports SFR by task suite. The SFR is 86.2% on synthetic tasks, 3.8% on UCI, and 18.9% on DSBench. We note that SFR and execution-success rate share the variable E (loud failures), so their correlation is partially mechanical. The $R^2 = 0.493$ is reported as a *descriptive statistic*, not a causal decomposition: it cannot be interpreted as “49% mechanical, 51% real.” A proper test would require a multinomial mixed-effects model with model, task difficulty, and suite as random effects, which we leave to future work.

4.4 Result 3: Taxonomy Validation

The final gold set combines an earlier 47-trace adjudicated subset with 103 newly sampled traces independently labeled by three annotators. On the new tranche, all annotators reproduce five blinded QC duplicates exactly, but raw agreement is only Fleiss’ $\kappa = .350$ for stage (95% CI [.235, .447]) and .590 for observability. The principal disagreement is whether misreading empty stdout constitutes Analytical-Plan or Output-Mismatch; 87 traces undergo consensus

Table 3: Comprehensive baseline results with infrastructure failure separation. SR = success rate (raw); SR^v = success rate among valid runs (excluding infra failures); Infra = runs with tokens=0 (API/parse/format failures); SFR = silent-failure rate among valid failures; Tok = mean tokens per valid run. “—” indicates all runs are infra failures (no valid data). DeepSeek-V3 is strongest on synthetic (88.4%) and UCI (94.2%). DSBench cells for DeepSeek-R1/Qwen3-max are n/a (API rate-limited).

Model	Suite	Runs	Infra	Valid	SR (%)	SR^v (%)	SFR (%)	Silent	Tok
GPT-4o	Synthetic	525	110	415	62.1	78.6	100.0	89	1,241
	UCI	69	0	69	91.3	91.3	100.0	6	1,543
	DSBench	600	0	600	3.3	3.3	38.3	222	12,792
GPT-4o-mini	Synthetic	525	0	525	73.9	73.9	39.4	54	3,341
	UCI	69	0	69	82.6	82.6	0.0	0	4,669
	DSBench	600	241	359	0.8	1.4	0.0	0	10,790
DeepSeek-V3	Synthetic	285	0	285	88.4	88.4	100.0	33	1,717
	UCI	69	0	69	94.2	94.2	0.0	0	2,410
	DSBench	600	600	0	0.0	—	—	0	0
DeepSeek-R1	Synthetic	285	285	0	0.0	—	—	0	0
	UCI	69	69	0	0.0	—	—	0	0
	DSBench	n/a (API rate-limited)							
Qwen3-max	Synthetic	285	214	71	18.2	73.2	100.0	19	770
	UCI	69	69	0	0.0	—	—	0	0
	DSBench	n/a (API rate-limited)							

Table 4: SFR by task suite and null-model analysis. R^2 is reported as a descriptive statistic, not a causal decomposition.

Suite	SFR (%)	Exec. Success (%)
Synthetic	86.2	45.5
UCI	3.8	73.6
DSBench	18.9	1.4
Null model: $R^2 = 0.493$ (descriptive only)		

adjudication. This moderate agreement is itself evidence that stage definitions require explicit operational rules.

The adjudicated distribution is Output-Mismatch 77, Runtime 39, Answer-Error 22, Code-Generation 12, and Analytical-Plan 0. The revised oracle-assisted classifier achieves 91.3% accuracy (95% CI [86.7, 95.3]) and 90.3% macro-F1 over observed classes. Its Code-Generation precision/recall are both 91.7%; stage and observability are scored independently.

4.5 Result 4: Zero Self-Correction

Across all 5,984 runs and all five models, the recovery rate is 0%: no agent ever corrected a silent failure without external intervention. We note that this is partly a consequence of the experimental protocol: silent failures produce no error signal, so the agent has no opportunity to detect and retry. Our recovery benchmark (Section 4.11) addresses this by testing feedback signals.

This zero-recovery finding implies that in the current setting without explicit correctness feedback or internal validation protocol, silent failures will not trigger automatic repair. In a multi-step pipeline, a silent failure at step k propagates to all downstream steps.

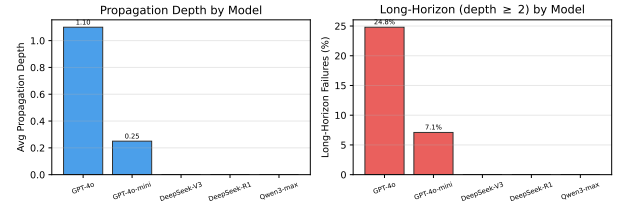


Figure 3: Propagation depth by model (baseline, valid runs only). GPT-4o exhibits the longest failure propagation (mean 1.10, 24.8% long-horizon). Other models show depth 0, which may reflect either immediate error detection or early trace termination due to infrastructure failures (see Table 3).

4.6 Result 5: Propagation Depth Analysis

Section 3.4 defines propagation depth. Here we analyse propagation depth by model, revealing a striking pattern (Figure 3). GPT-4o has a mean propagation depth of 1.10 steps with 24.8% of its failures reaching long-horizon status (depth ≥ 2)—far higher than any other model. GPT-4o-mini has a mean depth of 0.25 (7.1% long-horizon), while DeepSeek-V3, DeepSeek-R1, and Qwen3-max all have zero propagation depth.

We caution that zero propagation depth has two possible interpretations: (1) the failure was immediately detected by the agent (a loud failure at depth 0), or (2) the trace terminated early due to infrastructure failures (API errors, parse failures, or empty responses), leaving no steps to propagate through. Our enriched manifest (manifest_v3.csv) distinguishes these cases via the `is_infra_failure` field. Among valid (non-infrastructure) failures, GPT-4o’s longer propagation reflects its tendency to continue reasoning on a wrong intermediate result rather than crashing, making the failure harder

Table 5: Oracle-guided repair results. The full oracle rate (526 tasks) and the negative-control subset (56 tasks) use different samples; the no-op control confirms that oracle repair measures GT-code correctness, not step-specific causality.

Metric	Tasks	Rate (%)
Oracle repair (full)	349/526	66.3
<i>Negative control subset (56 tasks):</i>		
Oracle repair	36/56	64.3
No-op (GT code only)	36/56	64.3
Conservative TP	0/56	0.0

Table 6: Verifier ablation (McNemar test). *b*: baseline correct & verifier wrong; *c*: baseline wrong & verifier correct. Bold indicates $p < 0.05$.

Model	Base SR	Ver SR	Δ	<i>b/c</i>
GPT-4o-mini	73.9%	72.6%	-1.3	3/1
GPT-4o	62.1%	81.1%	+19.0	2/40
DeepSeek-V3	88.4%	87.7%	-0.7	2/1
Qwen3-max	18.2%	43.2%	+25.0	0/69

to localise. In contrast, DeepSeek-V3’s valid failures are concentrated at depth 0, meaning the error manifests in the final answer without intermediate propagation.

4.7 Result 6: Oracle-Guided Repair

Oracle-guided repair succeeds on 66% of failed tasks where ground-truth code is available (Table 5). This demonstrates *reparability*, not causal attribution: as the negative control in Section 5.3 shows, injecting the oracle solution produces identical results regardless of which step is replaced, because the ground-truth code independently computes the correct answer. We therefore report oracle repair as an upper bound on reparability rather than as evidence of step-level causal localisation. A full causal analysis with necessity/sufficiency testing is future work.

4.8 Result 7: Verifier Ablation

Table 6 reports the McNemar test results. The verifier significantly helps Qwen3-max (+24.9%, $p < 0.001$) and GPT-4o (+19.0%, $p < 0.001$), but is not significant for GPT-4o-mini, DeepSeek-V3, or DeepSeek-R1. Note that GPT-4o’s improvement was not detected by the paired t -test in our earlier analysis, highlighting the importance of using the correct statistical test for binary outcomes. The verifier’s benefit to Qwen3-max may partly reflect format/protocol fix-up rather than reasoning correction, since Qwen3-max’s low baseline performance is partly due to ReAct parser incompatibility.

A key finding is that the verifier’s effect is *model-dependent*: it helps models with low-to-moderate baseline performance (Qwen3-max at 18.2%, GPT-4o at 62.1%) but not the strongest model (DeepSeek-V3 at 88.4%) or the weakest (DeepSeek-R1 at 0%). This suggests that rule-based verification captures a “middle band” of errors—those where the model can reason about the problem but benefits from a structured check—but cannot rescue fundamentally broken traces (R1) or improve near-perfect ones (V3).

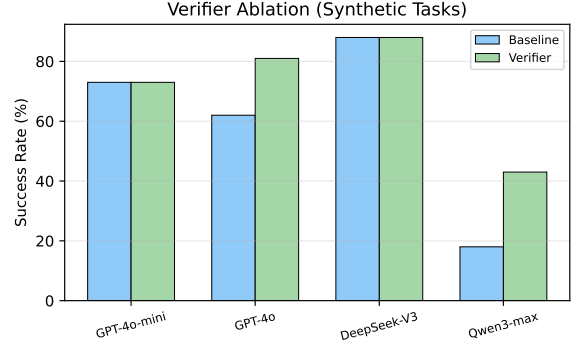


Figure 4: Verifier ablation. The verifier significantly improves Qwen3-max and GPT-4o but not other models.

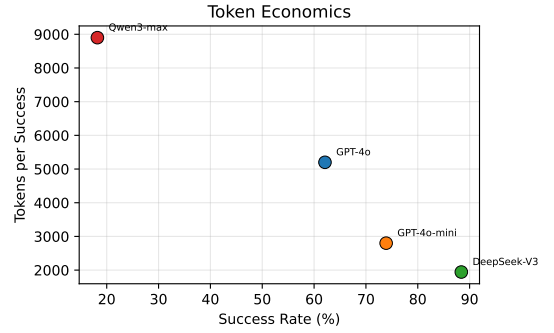


Figure 5: Tokens per success vs. success rate. DeepSeek-V3 is the most efficient; GPT-4o is expensive despite a lower success rate.

4.9 Result 8: Token Economics

Figure 5 plots tokens-per-success against success rate. DeepSeek-V3 is the most efficient (1,943 tokens/success); GPT-4o is the most expensive despite a lower SR. On DSBench, token costs escalate dramatically (10,790–12,792 tokens/run) due to complex multi-sheet workbooks, yet success rates remain near zero—a cost-effectiveness failure unique to real-world data.

4.10 Result 9: Harness Ablation

To verify that silent failures are not an artifact of our minimal ReAct harness, we compare three harnesses on both synthetic (30-task subset, 3 reps) and UCI (23 tasks, 3 reps) tasks: H1 (minimal ReAct, baseline), H2 (single-shot, one LLM call produces a complete script), and H3 (few-shot ReAct with one worked example). On synthetic, GPT-4o shows SFR of 100% (H1), 30% (H2), and 100% (H3); on UCI, the pattern is consistent at 100%, 31%, 100%. The harness significantly affects the failure distribution, so we report harness as a benchmark dimension rather than a nuisance factor. Details in Appendix C.

4.11 Result 10: Recovery Benchmark

To test whether feedback signals enable agents to recover from failures, we run a recovery benchmark with 6 conditions on 100 failed traces (80 silent + 20 loud) sampled from synthetic tasks:

- **C1 (No feedback):** Re-run the task with no additional signal.
- **C2 (Generic feedback):** “Your previous answer may be incorrect. Please retry.”
- **C3 (Logic hint):** “Check your logic carefully (e.g., count vs sum, groupby vs filter).”
- **C4 (Verification checklist):** “Verify your result by checking data types and aggregation function.”
- **C5 (Critic feedback):** GPT-4o-mini reviews the answer and provides critique.
- **C6 (Independent rerun):** “Solve using a completely different approach.”

The overall recovery rate is **15.7%** (94/600, 95% bootstrap CI [12.9, 18.5]%), a substantial improvement over the 0% recovery in the standard protocol. We analyse recovery at three levels: *attempt* (the agent produces any answer), *detection* (the agent produces a different answer than the original failed attempt, indicating it recognised a potential error), and *repair* (the new answer is correct). The attempt rate is high (97–100% across conditions), indicating that the agent always tries to respond. The detection rate (producing a different answer) varies by condition, and the repair rate is 15.7% overall. Critically, **loud failures recover at 36.7% (44/120, 95% CI [27.5, 45.0]%) while silent failures recover at only 10.4% (50/480, 95% CI [7.7, 13.1]%)**; the difference is highly significant ($\chi^2 = 48.1$, $p < 0.000001$), confirming that silent failures remain far harder to correct even with explicit feedback. No feedback condition significantly outperforms no-feedback (McNemar $p > 0.05$ for all conditions; per-condition 95% CIs overlap substantially: C1 [10.0, 24.0]%, C2 [10.0, 25.0]%, C3 [10.0, 25.0]%, C4 [8.0, 21.0]%, C5 [9.0, 22.0]%, C6 [8.0, 21.0]%), suggesting that the mere act of re-running with any prompt-level signal triggers partial recovery, but no single feedback type is dramatically more effective.

4.12 Result 11: Benchmark Tasks and Baselines

We define six potential benchmark tasks. Tasks B–D are the core diagnostic tasks; Tasks A, E, F are auxiliary tasks supported by the framework but not the primary focus:

- **Task A (Data-science task solving):** Given data and a question, produce the correct answer. This is the standard task; we report SR as the baseline metric.
- **Task B (Observability classification):** Given a trace, predict success / loud / silent. Metric: accuracy and macro-F1.
- **Task C (Failure-stage classification):** Predict one of the four stages observed in final gold: Output-Mismatch, Runtime, Answer-Error, or Code-Generation. Analytical-Plan remains in the ontology but has zero evaluation support. Metrics are accuracy, macro-F1, balanced accuracy, and per-class scores.
- **Task D (Originating-step localization):** Predict the first erroneous step. Metric: step-level accuracy and mean absolute error.
- **Task E (Verifier evaluation):** Given a trace without ground truth, detect whether the answer is correct. Metric: precision, recall, F1. Supported by the verifier ablation (Section 4.8).

Table 7: Failure-stage baselines on the final 150-trace human-adjudicated gold set. Macro-F1 is over the four observed classes. The rule classifier uses the reference path and is therefore oracle-assisted.

Baseline	Accuracy (%)	Macro-F1 (%)
Majority (Output-Mismatch)	51.3	17.0
Code length (< 200)	48.0	16.3
Answer presence	41.3	19.9
Observable exception	77.3	45.5
Rule classifier (oracle-assisted)	91.3	90.3
<i>Step localisation on 103 traces:</i>		
Random step (seed 2027)	35.9	—
Last code step	32.0	—

- **Task F (Repair):** Given a failed trace, generate a minimal fix. Metric: repair success rate. Supported by the recovery benchmark (Section 4.11) and oracle repair (Section 4.7).

Evaluation protocol. Table 7 scores leakage-free deterministic baselines on all 150 human-gold traces. The majority, code-length, answer-presence, and observable-exception methods are trace-only; the rule classifier additionally uses the reference path. Historical ML models trained to imitate binary auto-labels and legacy LLM judges are excluded because their features or targets do not match the final four-class task. For Task D, random- and last-step baselines are evaluated only on the 103 traces with adjudicated originating steps; neither is saturated (35.9% and 32.0%), reversing the earlier degenerate 100% result.

5 DISCUSSION

5.1 Why Strong Models Fail Silently

On synthetic trap tasks, the corrected SFR is high (86–100%) because these tasks are designed so that code runs but the answer is wrong. On real UCI data, SFR drops to 4%, and on DSBench to 19%. A null-model analysis ($R^2 = 0.493$, descriptive only) shows part of this variation is a mechanical consequence of execution-success differences: stronger models crash less, so their remaining failures are more likely silent. However, SFR and execution-success share the loud-failure count, so this correlation is partially mechanical by construction and cannot be decomposed into “mechanical” vs. “real” components. The practical implication is that improving code-generation ability alone will shift the failure distribution toward silent failures, but silent failures are most prevalent on tasks where code execution succeeds but reasoning fails.

5.2 Implications for Agent Deployment

The zero recovery rate (Section 4.5) has deployment implications. In the current setting without explicit correctness feedback, an agent will not self-correct silent failures, and wrong answers propagate through multi-step pipelines. However, our recovery benchmark (Section 4.11) shows that feedback signals enable 15.7% recovery, suggesting that external verification or in-loop feedback mechanisms can mitigate this issue. Our verifier ablation (Section 4.8)

Table 8: Negative control analysis. The no-op intervention (running GT code independently) produces identical results to the oracle intervention, confirming that oracle repair measures GT code correctness, not step-specific causal effect.

Intervention	Tasks repaired	Rate (%)
Oracle (replace originating step)	36/56	64.3
No-op (run GT code independently)	36/56	64.3
Conservative TP (oracle $\wedge$ $\neg$ noop)	0/56	0.0

further shows that simple rule-based verification helps weak models but not strong ones, indicating that effective verification likely requires model-specific calibration or learned verifiers.

5.3 Negative Control: Oracle Repair Revisited

To test whether the oracle repair rate (Section 3.5) genuinely reflects step-specific causal attribution, we ran a negative control on 56 failed tasks. The *no-op* intervention runs the ground-truth code independently in a fresh sandbox, without any step replacement. As Table 8 shows, the two rates are identical (64.3%), and the conservative true-positive rate is 0.0%. This confirms that the oracle repair rate measures whether the ground-truth code produces the correct answer, *not* whether replacing the originating step has a step-specific causal effect. We therefore report oracle repair as an upper bound on reparability rather than as evidence of root-cause localisation.

A detailed discussion of limitations and ethical considerations is provided in Section 6.

6 LIMITATIONS AND ETHICAL CONSIDERATIONS

Limitations. First, our minimal ReAct harness is controlled but not representative of production code-interpreter or AutoGen-style systems; the harness ablation confirms that failure distributions change with scaffolding. Second, 38.1% of the original runs are infrastructure failures, and several model-suite cells have no valid coverage; these cells must not be interpreted as model inability. Third, the final 150-trace gold set observes four of five taxonomy stages: Output-Mismatch (77), Runtime (39), Answer-Error (22), and Code-Generation (12), while Analytical-Plan has no support. The newly annotated 103-trace tranche has modest raw pre-adjudication stage agreement ($\kappa = .350$, 95% CI [.235, .447]) despite perfect within-annotator QC replication; this reflects a systematic boundary dispute resolved through consensus, not strong innate class separability. Fourth, the revised classifier is oracle-assisted because it compares traces with the reference solution path. Its 91.3% final-gold accuracy should not be interpreted as trace-only deployment performance, and its rules were refined after inspecting the gold errors; independent held-out validation is required. Finally, model aliases and the third-party API are not fully reproducible, and the remaining automatically labeled traces inherit classifier noise.

Ethical considerations. No personal data are used; the tasks derive from public UCI and DSbench/ModelOff resources or synthetic generators. The third-party endpoint may log prompts, so users should not submit sensitive data. Closed-model, English-language,

and Python-only coverage limits generalisability. Infrastructure failures are reported separately to prevent misleading model rankings. Public tasks and fixed synthetic seeds permit contamination or benchmark gaming; diagnostic metrics should therefore accompany raw success rates.

Generative AI usage. LLMs are experimental subjects and, in legacy analyses, auxiliary judges. The final 150 labels are human adjudications; LLM pre-labels are not treated as gold.

7 CONCLUSION

AgentFail provides full-trace diagnosis for data-science LLM agents, separating five failure stages from the independent question of whether a failure is observable. Across 5,984 runs, silent-failure rates remain strongly task-dependent and feedback recovers silent failures substantially less often than loud failures. The final 150-trace human gold set contains 77 Output-Mismatch, 39 Runtime, 22 Answer-Error, and 12 Code-Generation cases; Analytical-Plan remains uninstantiated. Raw stage agreement on the newly annotated tranche is modest ($\kappa = .350$), underscoring taxonomy difficulty, while consensus adjudication supports a revised oracle-assisted classifier with 91.3% accuracy and 90.3% macro-F1 over observed classes. Oracle repair matching its no-op control limits causal claims. Future work should validate the classifier on an independent gold set, populate Analytical-Plan naturally, and test production agent harnesses.

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A DATA CARD AND BENCHMARK DOCUMENTATION

A.1 Task Suite Descriptions

Synthetic trap tasks (95 tasks). 80 tasks are programmatically generated across 5 domains (tabular EDA, time-series, recommendation, statistical, text-log), with 16 instances per domain using a fixed seed (seed=16). Tasks share structural templates within each domain but use different generated data, producing 70/80 unique answers (87.5% uniqueness). Near-duplicate questions (identical first 100 chars) exist within domains (55/80) due to shared question structure, but the underlying data and expected answers differ. The remaining 15 tasks are DSBench questions (identified by dsbench_00000001 or dsbench_00000002 prefixes in the task ID) that are structurally similar to synthetic traps but originate from DSBench workbooks. Train/dev/test splits are by task ID, not by template, so the same template appears in all splits—this is intentional, as the goal is to evaluate failure diagnosis, not template memorisation.

UCI real-data tasks (23 tasks). Analytical questions on 6 public datasets (Iris, Titanic, Tips, MPG, Penguins, Flights) downloaded

Table 9: Task difficulty distribution by suite. Easy: $SR \geq 75\%$; Medium: $25\% \leq SR < 75\%$; Hard: $SR < 25\%$.

Suite	Tasks	Easy	Medium	Hard
Synthetic	95	19	63	13
UCI	23	0	21	2
DSBench	200	0	3	197

Table 10: Baseline success rates (%) by model and suite. “n/a” indicates API rate limiting prevented evaluation.

Model	Synthetic	UCI	DSBench
DeepSeek-V3	88.4	94.2	0.0
GPT-4o-mini	73.9	82.6	0.8
GPT-4o	62.1	91.3	3.3
Qwen3-max	18.2	0.0	3.0
DeepSeek-R1	0.0	0.0	n/a

at runtime from public repositories. Answers are computed deterministically from the real data. Tasks cover groupby aggregation, correlation, counting, and filtering.

DSBench real tasks (200 tasks). Real financial-analysis questions from ModelOff competitions [5], using original multi-sheet Excel workbooks. Answers are multiple-choice (A–I) or numeric. Tasks require understanding complex financial data structures.

A.2 Contamination and Template Analysis

The synthetic tasks are generated from 5 templates (one per domain), each producing 16 instances with different data. The answer distribution shows 70/80 unique answers (87.5%), with the most common answer appearing in only 4 tasks. This indicates low risk of answer-pattern memorisation. The train/dev/test split (198/42/43 tasks) is stratified by task ID across all suites; since the same template appears in all splits, models cannot exploit template-specific patterns. For DSBench, the tasks originate from public ModelOff competitions and may have appeared in LLM training data; we do not claim contamination-free evaluation for DSBench, only diagnostic trace analysis.

A.3 Task Difficulty Distribution

We characterise task difficulty by the baseline success rate aggregated across all five models. Table 9 reports the distribution. Synthetic tasks span the full difficulty range (19 easy, 63 medium, 13 hard); UCI tasks are uniformly medium-hard; DSBench tasks are almost entirely hard (197 of 200 tasks have $SR < 25\%$), reflecting the complexity of real financial data.

A.4 Expected Baseline Performance

Table 10 reports the expected baseline performance for each model and suite, providing reference points for future work. DeepSeek-V3 is the strongest on synthetic (88.4%) and UCI (94.2%) tasks. DSBench is challenging for all evaluated models (0–3.3%).

A.5 Data Schema

Each run in the enriched manifest (manifest_v3.csv, 32 fields) contains:

- `task_id`, `suite`: task identifier and suite (synthetic/uci/dsbench)
- `model`, `model_snapshot`, `provider`: model name, version info (rolling alias vs pinned snapshot), API provider
- `harness`, `variant`: agent harness type and experiment variant (baseline/verifier)
- `rep`: repetition index (0–2)
- `api_status`, `parser_status`, `execution_status`: infrastructure status
- `task_correct`: whether the final answer matches ground truth
- `observability`: success/silent/loud
- `failure_stage`, `failure_subcategory`: five-stage taxonomy label and subcategory
- `is_infra_failure`, `infra_subtype`: infrastructure failure flag and subtype (`api_error/unknown/none`)
- `originating_step`, `propagation_depth`: failure localisation and spread
- `is_silent`, `recovered`: silent failure flag and recovery status
- `tokens`, `tokens_input`, `tokens_output`, `cost_usd`: token usage and cost
- `final_answer`: agent’s final answer (truncated to 200 chars)
- `label_source`: rule (rule-based classifier) or unmatched
- `annotation_confidence`, `human_gold`, `adjudication_status`: human annotation fields populated for all 150 gold traces

Annotation uncertainty. The release contains 150 human-adjudicated traces. The new 103-trace tranche preserves three independent labels, five blinded QC duplicates per annotator, and consensus adjudications. Raw agreement is Fleiss’ $\kappa = .350$ for stage and $.590$ for observability; the revised oracle-assisted classifier achieves 91.3% accuracy against final gold. Remaining traces retain rule labels and should not be treated as equivalent to human gold.

Missing and duplicate runs. The manifest contains 5,984 runs. Some (task, model, variant) groups have fewer than 3 repetitions due to API failures or timeouts; these are retained with `api_status=error`. Duplicate runs (same task, model, variant, rep) are detected by the assertion checks and removed. Infrastructure-failed runs are retained in the manifest (for transparency) but excluded from valid-run analyses via the `is_infra_failure` filter.

A.6 Provenance and Reproducibility

All experiments use deterministic data generation (fixed seed=16 for synthetic tasks) and temperature=0 LLM calls. The framework includes a MockLLM backend that reproduces the full pipeline without API cost. Real-model experiments use the ChatAnywhere OpenAI-compatible endpoint (API calls: June–July 2026). Total API cost: \$80. Note that four of five model identifiers (gpt-4o, gpt-4o-mini, deepseek-chat, deepseek-reasoner) are rolling aliases whose underlying model may change; only qwen3-max-2026-01-23 is a pinned snapshot. Repetition agreement under temperature 0 is 89.9%, with disagreement attributed to server-side non-determinism.

The repository includes a `requirements.txt` for dependency pinning and MD5 checksums for key artifacts (manifest_v3.csv: 29ceae8..., manifest.csv: 6e2f809...). A Docker image is planned for the camera-ready version.

A.7 Answer Evaluation Details

Answers are evaluated as follows: (1) *Numeric answers*: compared with a relative tolerance of 10^{-3} (e.g., $\text{abs}(\text{agent} - \text{gt}) / \max(\text{abs}(\text{gt}), 1e-9) < 1e-3$). (2) *String answers*: normalised by stripping whitespace, converting to lowercase, and removing punctuation before exact match. (3) *Multiple-choice answers* (DSBench): matched on the letter (A–I) only, ignoring surrounding text. (4) *Tuple answers* (e.g., “(species, value)”): each component is evaluated independently; both must match. The evaluator code is in `agentfail/benchmark/tasks.py`.

A.8 Known Issues

- DeepSeek-R1 produces no valid traces on any suite (100% infrastructure failure); its results should not be interpreted as model inability.
- DeepSeek-V3 on DSBench: all 600 runs are infrastructure failures (tokens=0), likely due to Excel parsing issues in the sandbox.
- Qwen3-max on UCI: all 69 runs are infrastructure failures.
- Human adjudication covers 150 failures; the revised oracle-assisted classifier reaches 91.3% accuracy but requires independent held-out validation because its rules were refined after gold error analysis.
- DSBench tasks require a separate download and may be present in LLM training data.

A.9 Licensing and Maintenance

The code is released under an open-source license (MIT). The synthetic task generators are self-contained and require no external data. UCI datasets are public domain. DSBench tasks are used under their original license [5]; users must obtain DSBench separately. Raw LLM responses are not redistributed; users must re-run experiments to obtain them. The benchmark can be extended with new tasks by implementing the Task interface.

Maintenance plan. The repository is maintained by the authors. Issues and pull requests are accepted via GitHub. The maintainers commit to:

- Responding to issues within 30 days.
- Releasing compatibility patches for new LLM API versions.
- Preserving backward compatibility of the manifest schema across minor versions.
- Archiving major versions as tagged releases.

Versioning. The benchmark follows semantic versioning (MAJOR.MINOR.PATCH). The current release is v1.0.0. Breaking changes to the manifest schema or task definitions increment MAJOR; new tasks or metrics increment MINOR; bug fixes increment PATCH. Each release is tagged on GitHub with a corresponding Zenodo DOI for long-term archival.

Intended use cases. The benchmark is designed for: (1) evaluating LLM-agent failure modes on data-science tasks; (2) comparing diagnostic metrics (SFR, propagation depth, recovery rate) across

Table 11: Propagation depth distribution (all 3,940 failures).

Depth	Failures	Percentage
0	3,350	88.6%
1	169	4.5%
2	80	2.1%
3	39	1.0%
4	64	1.7%
5	45	1.2%
6	20	0.5%
7	14	0.4%
≥ 2 (long-horizon)	262	6.9%

models; (3) testing new verifier or repair strategies. It is *not* intended for: (1) training LLMs (tasks have deterministic answers); (2) production deployment evaluation (the minimal ReAct harness differs from production systems); (3) cross-domain generalisation claims (the three suites cover data analysis, not all of data science).

A.10 Propagation Depth Analysis

Table 11 reports the propagation depth distribution across all 3,940 failures. The average depth is 0.30 steps, meaning most failures are detected immediately. However, 262 failures (6.9%) propagate to depth ≥ 2 , indicating long-horizon failure spread where the agent continues building on a wrong intermediate result.

B ANNOTATION PROTOCOL AND RESULTS

The final evaluation set contains 150 human-adjudicated failed traces: an earlier 47-trace tranche and 103 newly sampled traces. Three annotators independently label the new tranche while blinded to model and system predictions; each reproduces five duplicate QC traces exactly. Raw Fleiss' agreement is $\kappa = .350$ for stage (95% CI [.235, .447]) and .590 for observability. Stage full agreement is 27.2%; 87 traces undergo consensus adjudication. The final distribution is Output-Mismatch 77, Runtime 39, Answer-Error 22, Code-Generation 12, and Analytical-Plan 0. Raw labels, QC mappings, adjudications, and gold labels are released.

C HARNESS ABLATION DETAILS

Table 12 reports the full harness ablation results on synthetic tasks. We also ran the same three harnesses on all 23 UCI tasks (3 repetitions each), with consistent results: GPT-4o shows SFR of 100% (H1), 31% (H2), and 100% (H3) on UCI, matching the synthetic pattern (100%/30%/100%). This confirms silent failures are not an artifact of the minimal ReAct design across both task types.

D RECOVERY BENCHMARK DETAILS

The recovery benchmark (Section 4.11) tests 100 failed traces (80 silent + 20 loud) under 6 feedback conditions, yielding 600 runs. Overall recovery rate is 15.7%. The key finding is the asymmetry between failure types: loud failures recover at 36.7% (44/120) while silent failures recover at only 10.4% (50/480), confirming that silent failures remain substantially harder to correct even with explicit feedback signals.

Table 12: Harness ablation. Three harnesses tested on 30 synthetic tasks $\times$ 3 repetitions. H1 = minimal ReAct (baseline); H2 = single-shot (one LLM call, complete code); H3 = few-shot ReAct (ReAct + 1 worked example). Silent failures persist across all three harnesses for GPT-4o, confirming they are not an artifact of the minimal ReAct design. DeepSeek-V3 cells are partial due to long API latencies.

Model	Harness	n	SR (%)	SFR (%)	Silent
GPT-4o	H1: ReAct	90	86.7	100.0	12
	H2: Single-shot	90	44.4	30.0	15
	H3: Few-shot ReAct	90	82.2	100.0	16
GPT-4o-mini	H1: ReAct	90	70.0	0.0	0
	H2: Single-shot	90	53.3	28.6	12
	H3: Few-shot ReAct	90	85.6	0.0	0
DeepSeek-V3	H1: ReAct	11	100.0	0.0	0
	H2: Single-shot	17	82.4	0.0	0

E REPRODUCIBILITY

All experiments use deterministic data generation (fixed seeds for synthetic tasks) and temperature-0 LLM calls. The framework includes a deterministic MockLLM that reproduces the full pipeline without API cost. Real-model experiments use the ChatAnywhere OpenAI-compatible endpoint. Total API cost: \$80. Code and data: https://github.com/llmnjust-afk/KDD2027_Work2.

F DATA AVAILABILITY

All code, data, and traces are available at https://github.com/llmnjust-afk/KDD2027_Work2. The release includes the 32-field manifest, the 150-trace human gold set with raw labels and adjudications, the 927-run relaxed-sandbox extension, deterministic baseline scripts, and dependency specifications. Synthetic tasks are self-contained; UCI data download at runtime, while DSBench requires its separately licensed source [5].